

Muhamad Andung Muntaha

andung07@gmail.com
+6281380260278

Jakarta, Indonesia
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Solution Architect — AI, IoT, & Cloud. Solution Architect with 12+ years designing and delivering end-to-end technology solutions spanning embedded systems, IoT platforms, cloud-native architectures, and web and mobile application development. Proven track record translating business and client requirements into scalable technical architectures — from custom hardware and real-time firmware to Kubernetes-based distributed systems and client-facing web and mobile products in the cloud. Comfortable operating at both the whiteboard and the codebase: leads architecture reviews, defines technology roadmaps, and acts as trusted technical advisor to clients and executive stakeholders through complex digital transformation initiatives. Experience spanning oil & gas, automotive/fleet, wearables, edge AI, and enterprise IT.

Core Competencies

- **Solution & Systems Architecture:** End-to-end IoT/edge-to-cloud architecture, distributed systems design, microservices, digital twins, technology roadmapping
- **Client & Stakeholder Engagement:** Consultative selling, pre-sales technical advisory, requirements discovery, executive communication, business case development
- **Platforms & Infrastructure:** Kubernetes, Docker, Cloud deployment, CI/CD (Jenkins), observability (ELK, Grafana), edge computing, embedded Linux (Yocto)
- **Embedded & Real-Time Systems:** Zephyr, FreeRTOS, RT-Thread, MbedOS, STM32/NRF52/ESP32, CAN/BLE/LoRa/MQTT protocol design
- **Industrial IoT & Control Systems:** SCADA integration, OT/IT convergence, closed-loop control systems, Modbus, OPC-UA, IIoT edge platforms (Agora), process simulation & digital twin integration (Symmetry, Pipesim)
- **Languages:** C, C++, C#, Go, Python, Java, JavaScript
- **Security & Governance:** Two-way TLS authentication design, ISC2 SSCP, CCNA, ITIL Foundation

Work Experience

- **XLabs Teknologi Indonesia — Founder** Jakarta, Indonesia
Mar 2025 - present
Full-time
 - Architect and deliver R&D and consulting engagements for enterprise clients adopting AI and IoT, scoping solutions from embedded edge devices through cloud data pipelines.
 - Architected and launched GymSense, an AI-powered fitness tracking engine using real-time computer-vision-based movement analysis.
 - Architected and launched Movaja, a mobile and web fitness application delivering pro-led workout programs with real-time on-device AI form scoring, personalized training plans, and a corporate wellness offering.
- **Netra Teknologi Nusantara — Chief Technology Officer** Jakarta, Indonesia
Oct 2023 - Feb 2025
Full-time
 - Acted as lead solution architect and trusted technical advisor for enterprise clients, translating business goals into tailored IoT architectures spanning custom embedded hardware and Kubernetes-based cloud systems.
 - Defined end-to-end technology stack decisions to ensure solutions were robust, scalable, and aligned with industry best practice, from device firmware to distributed cloud services.
 - Owned business development and consultative-selling motion, guiding clients through digital transformation roadmaps as the primary technical point of contact.
 - Built and mentored high-performing cross-disciplinary engineering teams to execute on architectural vision.
- **Schlumberger — Industrial IoT Engineer** Jakarta, Indonesia
June 2022 - Oct 2023
Full-time
 - Architected Indonesia's first commercial Industrial IoT solution: an edge-deployed, closed-loop machine-learning gas-lift optimization system, increasing oil production by 5 BOPD.
 - Designed real-time well-monitoring architecture integrating SLB's Agora IIoT platform with Avalon Lift Surveillance Software across distributed field assets.
 - Designed and integrated a digital twin architecture for a gas power plant, bridging existing SCADA systems with SLB's Symmetry process-simulation software.
 - Designed a digital twin solution for an oil pipeline network, integrating SCADA, data pipelines, and SLB's Pipesim flow-assurance software.
- **Blue Bird Group — IoT Software Lead** Jakarta, Indonesia
May 2021 - June 2022
Full-time
 - Architected an end-to-end IoT platform deployed across 20,000 vehicles, spanning embedded firmware, mobile/embedded Android, and cloud backend.

- Designed the system architecture connecting in-vehicle CAN bus data, embedded firmware (RT-Thread RTOS / STM32), and a Go-based microservices backend deployed on Kubernetes.
- Defined the architecture for a fleet-wide OTA update and device-management control center.
- Led a 4-engineer development team; designed a two-way TLS authentication architecture to secure device-to-cloud communication.

- **SoleMetrix — Embedded Software Engineer** Singapore (Remote)
Full-time June 2020 - May 2021
 - Designed the end-to-end embedded architecture (Zephyr RTOS) for a sports-technology wearable, including a custom UWB ranging protocol achieving 5cm precision.
 - Designed the BLE peripheral API and device driver architecture, plus a data-compression scheme for efficient sensor streaming to host systems.
- **Nodeflux — Embedded Software Lead** Jakarta, Indonesia
Full-time August 2019 - May 2020
 - Architected the embedded Linux platform (Yocto-based), OTA update strategy, and remote-management design for an edge AI computing platform, leading a team of three engineers.
- **Scania Research and Development — Thesis Research in Real-Time Systems** Stockholm, Sweden
Internship January 2019 - June 2019
 - Designed a centralized logging and monitoring architecture that reduced hardware-in-the-loop simulator troubleshooting time.
 - Introduced formal schedulability analysis into the system design process to validate real-time timing requirements.
- **Freelance — Embedded Software Developer** Stockholm, Sweden
Part-time August 2018 - December 2018
 - Client: Techbros GmbH Germany - Developed a cellular network automatic testing device, developed using Qt on embedded linux.
 - Client: Ontel AB Stockholm - Designed and implemented bike sharing IoT hardware prototypes in Arm MbedOS, satisfying low-power requirements and unreliable network constraints.
- **KTH MIST Student Satellite Project — Embedded Software Developer** Stockholm, Sweden
Project March 2018 - December 2018
 - Achieved a significant milestone in the end-to-end satellite communication architecture by implementing a MISRA-C compliant radio communication API.
 - Minimized functional testing time by introducing a radio link communication simulator.
 - Introduced a proof-of-concept in processing experiment data by extending the ground station software.
- **Schlumberger — IT Engineer & Application Developer** Jakarta, Indonesia
Full-time November 2013 - June 2017
 - Delivered \$5.5M in cost savings by architecting and building 15 SharePoint applications, 2 web applications, and 2 mobile applications that transformed internal business processes.
 - Won the 2017 Transformation Award for architecting a web/mobile solution that improved Account Receivable turnover.
 - Recognized as best IT engineer based on user satisfaction feedback and performance maintaining IT service quality and availability.

Education

- **KTH Royal Institute of Technology** Stockholm, Sweden
M.Sc. Embedded Systems, minor: Embedded Software, GPA: 3.72 / 4.00 August 2017 - June 2019
 - Related courses: Embedded Systems, Embedded Software, Software Reliability, Compiler and Execution Environment, Networked Systems Security, Modern Method in Software Engineering.
 - Thesis: Non-intrusive Logging and Monitoring System of A Parameterized Hardware-in-the-loop Real-Time Simulator.
- **10 Nopember Institute of Technology** Surabaya, Indonesia
B.Eng. Electrical Engineering, minor: Computer Engineering, GPA: 3.35 / 4.00 August 2009 - July 2013
 - Related courses: Embedded Systems, Computer Architecture, Data Communication, Computer Networks, Computer Vision, Algorithm & Data Structure, Artificial Intelligence.
 - Thesis: Self-localization of Humanoid Robot Inside Soccer Field Using Triangulation and Particle Filter Methods.

Industry Certifications

- Cisco Certified Network Associate Routing and Switching (CCNA) - 2016
- ITIL Foundation - IT Service Management - 2015
- Microsoft Certified IT Professional - 2015
- ISC2 System Security Certified Practitioner (SSCP) - 2014